

TED (15) – 4033

(REVISION – 2015)

FOURTH SEMESTER DIPLOMA EXAMINATION IN ELECTRICAL AND ELECTRONICS
ENGINEERING

MODEL QUESTION

ELECTRICAL ESTIMATING AND COSTING

(Maximum marks: 100)

Time: 3 Hrs

Part A

(Maximum marks: 10)

I. Answer all questions in one or two sentences. Each question carries 2 marks.

1. State Lambert's cosine law.
2. What are the materials used for the insulation of wiring cables.
3. Write the full specification of minimum size of cable for light wiring.
4. List out the insulators used in 11kV OH distribution lines.
5. Name the different types of 11kV distribution substation.

Part B

(Maximum marks: 30)

II. Answer any five of the following questions. Each question carries 6 marks.

1. Draw the Sodium Vapour lamp and identify the components.
2. List out 6 internal wiring rules
3. Describe the function of earthing and name the different types of earthing
4. Select suitable main switch, control switch and cable for a 400 v, 50 Hz, 3 phase 10 HP squirrel cage induction motor working at 0.8 pf and efficiency at 80%.
5. List out 6 main components of an 11 kV distribution pole
6. Estimate the list of materials for standard plate earthing.
7. Write short note on
 - a. Lightning arrester
 - b. AB switch
 - c. Earthing of distribution line

Part C

(Maximum marks: 60)

(Answer one full question from each unit. Each full question carries 15 marks.)

Unit - I

- III. Estimate the number and wattage of lamps which should be required to illuminate a workshop of 70x15m by means of lamp mounted 5m above the working plane. The average illumination required is about 100 lux, coefficient of utilization factor is 0.4, luminous efficiency 16 lumens/watt. Assume space height ratio as unity and candle power depreciation of 20%. Draw the layout and mark the disposition of lamps.

(15)

OR

- IV. A drawing hall of 30mx15m with ceiling height of 5m is to be provided with a general illumination of 120 lux. Taking coefficient of utilization of 0.5 and depreciation factor as 1.4, luminous efficiency of 80W fluorescent lamp is 40 lumens/watt. Determine
- Number of lamps required
 - Mounting height
 - Space height ratio
 - Layout of disposition of lamps

(15)

Unit -II

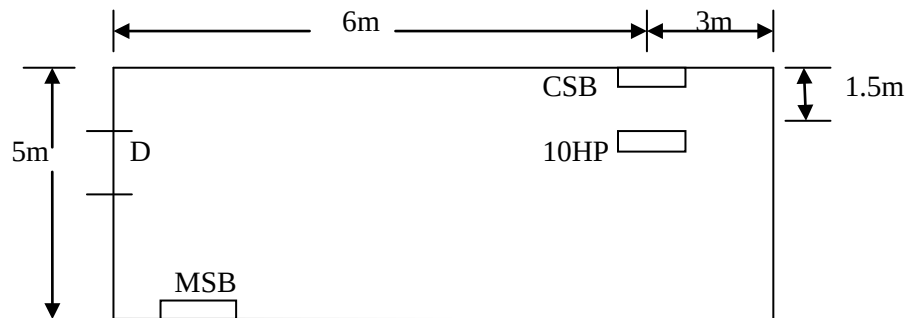
- V. a) List out different types of wiring cables with full specifications. (7)
 b) Select main switch, distribution board and number sub circuits for an installation having 8 numbers fans , 10 nos fluorescent tubes and 5nos ordinary sockets (8)

OR

- VI. Estimate a quantity of materials required with approximate cost for wiring in open conduit system as per internal wiring rules for a hall of size 10m x 5m x 3m. Main switch is placed near the door. The door is in the middle portion of the western side of the hall. (15)

UNIT-III

- VII A 10Hp, 400V, three phase 50Hz induction motor is to be installed in a workshop as per the given layout. Prepare the estimate of materials required with approximate cost. Take power factor as 0.8 and efficiency of motor as 80%.



(15)

OR

- VIII. Select main switch, control switches and cables for an industry having two 15HP motors and one 10 HP motors working at 0.8 pf and efficiency 80%.

(15)

Unit IV

- IX. A 1 km long OH LT three phase 50Hz line is to be erected from a 125KVA pole mounted substation using PCC poles of 8m length. If the span between the adjacent poles is 50m and using 6/1x2.59mm ACSR conductor for three phases and 6/1x2.11mm ACSR conductor as neutral. Make a list of materials required.

(15)

OR

- X. List out the major components of 400KVA 11/0.4kV Pole mounted substation.

(15)